

Week 4: Capacitors, Inductors and Transients

Energy storage, DC steady state, first-order (RC, RL) and second-order (RLC) transients

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 13 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §3.1–3.2 and §4.1–4.4

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

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Contents

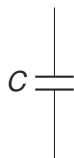
PART 1: Lecture notes	4–53
Energy storage, symbols	4–5
Capacitors	6–9
Inductors	10–12
Series and parallel C and L	13–15
DC steady state and stored energy	16–21
First-order transients: the recipe, RC circuits	22–29
RL circuits, the inductor kick, Thévenin + τ	30–33
Second-order (RLC) transients	34–49
Traps and exam tips	50–51
Summary	52–53
PART 2: Final check (8 practice problems)	54–62
PART 3: Answer sheet (full solutions)	63–73

How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

Energy storage elements



capacitor



inductor

A resistor burns energy. A capacitor and an inductor **store** it and give it back later.

Capacitor: energy in an electric field (charge on two plates).

Inductor: energy in a magnetic field (current in a coil).

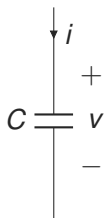
Why this changes everything

Their current–voltage laws have a **derivative**. So circuits with C or L are described by differential equations, and their response takes time: the **transient**.

Symbols for this week

Symbol	Unit	Meaning
C	farad (F)	capacitance. Real values: pF to mF (μ F is common)
L	henry (H)	inductance. Real values: μ H to H (mH is common)
τ	second (s)	time constant, RC or L/R
$x(0^-)$		value just before the switch moves
$x(0^+)$		value just after the switch moves
$x(\infty)$		final value, once everything has settled
α, ω_0	1/s, rad/s	damping and resonant frequency (second order)

Capacitor



$$q = Cv \quad \text{[on formula sheet]}$$

$$i = C \frac{dv}{dt} \quad \text{[on formula sheet]}$$

$$W = \frac{1}{2} Cv^2 \quad \text{[on formula sheet]}$$

$$C = \frac{\epsilon A}{d} \quad \text{parallel plates: area } A, \text{ gap } d$$

The two facts that solve most problems

1. **DC steady state:** v is constant, so $i = 0$. A capacitor is an **open circuit**.
2. **Continuity:** the voltage across a capacitor **cannot jump**. $v_C(0^+) = v_C(0^-)$.

Where the capacitor and inductor laws come from

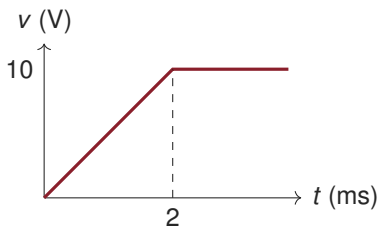
Capacitor: charge is proportional to voltage, $q = Cv$, and current is the rate charge arrives, $i = dq/dt$. Together: $i = C dv/dt$.

Current flows only while v is **changing**. Constant v means no current (open at DC), and a jump in v would need infinite current, so v_C can't jump.

Inductor: flux linkage is proportional to current, $\lambda = Li$, and Faraday says $v = d\lambda/dt$. Together: $v = L di/dt$.

Voltage appears only while i is **changing**. Constant i means no voltage (wire at DC), and i_L can't jump.

Try it: Example 1, capacitor



A $4\ \mu\text{F}$ capacitor has this voltage.

- (a) Current for $0 < t < 2$ ms?
- (b) Current after 2 ms?
- (c) Energy stored at $t = 3$ ms?

your work here ↓

Example 1: solution

(a) $\text{slope} = \frac{10 \text{ V}}{2 \text{ ms}} = 5000 \text{ V/s}$, $i = 4 \mu\text{F} \times 5000 = 20 \text{ mA}$

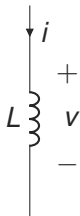
(b) v is flat, slope = 0: $i = 0$

(c) $W = \frac{1}{2}(4 \times 10^{-6})(10)^2 = 0.2 \text{ mJ}$

Engineering shortcut

Current in a capacitor = $C \times$ (slope of its voltage graph). Flat voltage means zero current, and a steep ramp means a big current.

Inductor



$$v = L \frac{di}{dt}$$

[on formula sheet]

$$W = \frac{1}{2} L i^2$$

[on formula sheet]

$$L = \frac{\mu N^2 A}{\ell}$$

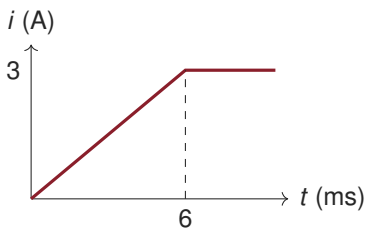
coil: N turns, area A , length ℓ

The mirror image of the capacitor

- DC steady state:** i is constant, so $v = 0$. An inductor is a **short circuit** (a wire).
- Continuity:** the current through an inductor **cannot jump**. $i_L(0^+) = i_L(0^-)$.

Shortcut: swap $C \leftrightarrow L$, $v \leftrightarrow i$, open $\leftrightarrow$ short. Learn one element and you know the other.

Try it: Example 2, inductor



A 2 mH inductor carries this current.

- (a) Voltage for $0 < t < 6$ ms?
- (b) Voltage after 6 ms?
- (c) Energy stored after 6 ms?

your work here ↓

Example 2: solution

(a) $\text{slope} = \frac{3 \text{ A}}{6 \text{ ms}} = 500 \text{ A/s}, \quad v = 2 \text{ mH} \times 500 = 1 \text{ V}$

(b) constant current, slope 0: $v = 0$

(c) $W = \frac{1}{2}(2 \times 10^{-3})(3)^2 = 9 \text{ mJ}$

Trap

A sudden jump in inductor current would need infinite voltage. That's why opening a switch on a coil makes a spark (see Example 7).

Series and parallel C and L

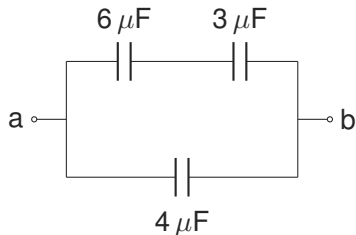
[on formula sheet]

	Series	Parallel
Inductors (like R)	$L_{eq} = L_1 + L_2 + \dots$	$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots$
Capacitors (opposite!)	$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$	$C_{eq} = C_1 + C_2 + \dots$

Trap

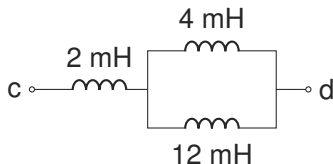
Capacitors combine the **opposite** way to resistors. Parallel capacitors **add**. Two in series use product over sum.

Try it: Example 3, combining C and L



(a) capacitors between a and b

Find C_{ab} and L_{cd} .



(b) inductors between c and d

your work here ↓

Example 3: solution

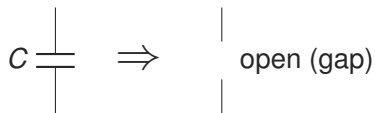
(a) $6\ \mu$ and $3\ \mu$ in series: $\frac{6 \times 3}{6 + 3} = 2\ \mu\text{F}$

that branch in parallel with $4\ \mu\text{F}$: $2 + 4 =$ $C_{ab} = 6\ \mu\text{F}$

(b) $4\ \text{mH} \parallel 12\ \text{mH}$: $\frac{4 \times 12}{16} = 3\ \text{mH}$

in series with $2\ \text{mH}$: $3 + 2 =$ $L_{cd} = 5\ \text{mH}$

DC steady state: C open, L short



open (gap)

After a DC source has been on “for a long time”, nothing changes anymore:

C $\rightarrow$ open circuit ($i_C = 0$)

L $\rightarrow$ short circuit ($v_L = 0$)



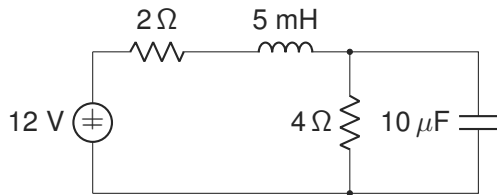
short (wire)

Then it's a plain resistive circuit (week 1–3 methods).

Engineering shortcut

“Find the energy stored” at DC: replace C with a gap and L with a wire, solve for v_C and i_L , then use $\frac{1}{2} C v_C^2$ and $\frac{1}{2} L i_L^2$.

Try it: Example 4, DC steady state and energy



The source has been on for a long time.

Find i_L , v_C , and the energy stored in each element.

your work here ↓

Example 4: solution

Replace $L \rightarrow$ wire, $C \rightarrow$ gap: one loop, 12 V, $2 + 4 = 6 \Omega$

$$i_L \quad 12/6 = \boxed{2 \text{ A}}$$

$$v_C \quad \text{same as across the } 4 \Omega: 2 \times 4 = \boxed{8 \text{ V}}$$

$$W_L \quad \frac{1}{2}(5 \times 10^{-3})(2)^2 = \boxed{10 \text{ mJ}}$$

$$W_C \quad \frac{1}{2}(10 \times 10^{-6})(8)^2 = \boxed{0.32 \text{ mJ}}$$

Trap

$v_C = 8 \text{ V}$, not 12 V: the capacitor sits across the 4Ω only, and the 2Ω takes its share.

Writing the differential equation

[NOT on formula sheet: memorize]

Use KVL or KCL, then substitute $i_C = C dv_C/dt$ or $v_L = L di_L/dt$:

Series RC (source V_s) $V_s = RC (dv_C/dt) + v_C \Rightarrow \tau (dv_C/dt) + v_C = V_s, \tau = RC$

Series RL $V_s = L (di_L/dt) + Ri_L \Rightarrow \tau (di_L/dt) + i_L = V_s/R, \tau = L/R$

Series RLC $LC (d^2v_C/dt^2) + RC (dv_C/dt) + v_C = V_s$

Standard first-order form

$\tau (dx/dt) + x = K f(t)$: τ = time constant, K = DC gain, final value $x(\infty) = Kf$. The week 4 formula is the solution of this equation.

Try it: Example 4b, write the equation

A 12 V source charges $C = 5 \mu\text{F}$ through $R = 4 \text{ k}\Omega$.

Write the differential equation for v_C in standard form. Identify τ and the DC gain, and give $v_C(\infty)$.

your work here ↓

Example 4b: solution

KVL $12 = RC \frac{dv_C}{dt} + v_C$

Numbers

$$0.02 \frac{dv_C}{dt} + v_C = 12$$

Read off

$\tau = 20 \text{ ms}$, DC gain $K = 1$, $v_C(\infty) = 12 \text{ V}$

Shortcut: in steady state $d/dt = 0$, so set the derivative to zero and read $x(\infty)$ straight from the equation.

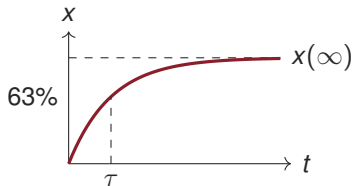
First-order transients: one formula

[NOT on formula sheet: memorize]

$$x(t) = x(\infty) + [x(0^+) - x(\infty)] e^{-t/\tau}$$

x = any voltage or current in a circuit with **one C** or **one L**.

$$\tau = R_{Th} C \quad \text{or} \quad \tau = \frac{L}{R_{Th}}$$



R_{Th} in τ

R_{Th} is the resistance **seen by the C or L** with independent sources killed, in the circuit **after** the switch moves.

Reading the formula term by term

$$x(t) = \underbrace{x(\infty)}_{\text{where it ends}} + \underbrace{[x(0^+) - x(\infty)]}_{\text{the gap at the start}} \underbrace{e^{-t/\tau}}_{\text{shrinks from 1 to 0}}$$

At $t = 0^+$: $e^0 = 1$, so $x = x(0^+)$ ✓. As $t \rightarrow \infty$: $e^{-\infty} = 0$, so $x = x(\infty)$ ✓.
In between, the gap closes smoothly.

After one τ : $e^{-1} = 0.368$, so 36.8% of the gap is left: 63.2% of the change is done.

Units: $\Omega \times F = \frac{V}{A} \cdot \frac{C}{V} = \frac{C}{C/s} = s$, and H/Ω is also seconds. So τ really is a time.

Charging ($0 \rightarrow 10 \text{ V}$) and discharging ($8 \text{ V} \rightarrow 0$) are the **same** formula: only $x(0^+)$ and $x(\infty)$ change.

How to find the R_{Th} in τ

- 1 Use the circuit **after** the switch has moved.
- 2 Take the capacitor (or inductor) out, leaving its two terminals open.
- 3 Kill every independent source: $V \rightarrow$ wire, $I \rightarrow$ gap.
- 4 Find the equivalent resistance between those two terminals (week 1 methods).
- 5 $\tau = R_{Th}C$ or $\tau = L/R_{Th}$.

Example 6: after the switch opens, the 12 V source and its 4 k are disconnected. From the capacitor's terminals only the 8 k is left, so $\tau = 8 \text{ k} \times 25 \mu\text{F} = 0.2 \text{ s}$.

Trap: it's the resistance **seen by the C or L**, not the total resistance in the circuit, and not the one before the switch moved.

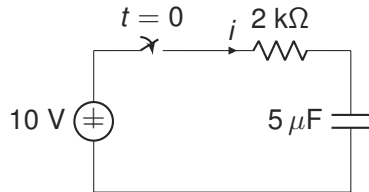
The recipe, and how fast is “fast”

- 1 $t = 0^-$: DC steady state of the old circuit: $v_C(0^-)$ or $i_L(0^-)$
- 2 Continuity: $v_C(0^+) = v_C(0^-)$,
 $i_L(0^+) = i_L(0^-)$
- 3 $t = \infty$: DC steady state of the new circuit: $x(\infty)$
- 4 τ from R_{Th} of the new circuit
- 5 Plug into the formula

t	% of the change done
1τ	63.2%
2τ	86.5%
3τ	95.0%
5τ	99.3% (“done”)

Shortcut: solve for v_C or i_L first (they’re continuous), then get anything else with KVL/KCL at that moment.

Try it: Example 5, RC charging



The capacitor starts uncharged. The switch closes at $t = 0$.

- (a) τ (b) $v_C(t)$ (c) v_C at $t = 10\text{ ms}$
(d) $i(0^+)$

your work here ↓

Example 5: solution

Initial $v_C(0^+) = v_C(0^-) = 0$

Final $C \rightarrow$ gap, no current, so $v_C(\infty) = 10 \text{ V}$

(a) $\tau = RC = (2000)(5 \times 10^{-6}) = 10 \text{ ms}$

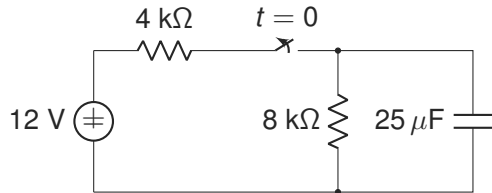
(b) $v_C(t) = 10(1 - e^{-t/0.01}) \text{ V}$

(c) $t = \tau: 10(1 - e^{-1}) = 6.32 \text{ V}$

(d) at 0^+ the C still has 0 V, like a wire: $i = 10/2k = 5 \text{ mA}$

Shortcut: at $t = 0^+$ an uncharged capacitor acts like a **wire**, and an inductor with zero current acts like a **gap**.

Try it: Example 6, switch opens



The switch was closed for a long time and opens at $t = 0$.

Find $v_C(0^+)$, τ , $v_C(t)$, and v_C at $t = 0.1$ s.

your work here ↓

Example 6: solution

$$t = 0^- \quad C \rightarrow \text{gap: divider, } v_C = 12 \cdot \frac{8}{4 + 8} = 8 \text{ V}$$

$$0^+ \quad \text{continuity: } v_C(0^+) = 8 \text{ V}$$

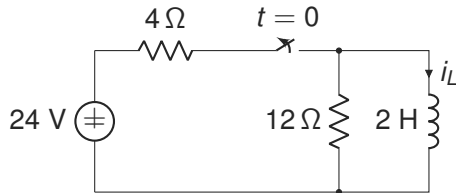
Final source disconnected, C discharges: $v_C(\infty) = 0$

$$\tau \quad C \text{ sees only the } 8 \text{ k: } \tau = (8000)(25 \times 10^{-6}) = 0.2 \text{ s}$$

$$v_C(t) \quad v_C(t) = 8 e^{-5t} \text{ V}$$

$$t = 0.1 \quad 8 e^{-0.5} = 4.85 \text{ V}$$

Try it: Example 7, the inductor kick



The switch was closed a long time and opens at $t = 0$.

Find $i_L(t)$ and the inductor voltage at $t = 0^+$.

your work here ↓

Example 7: solution

$t = 0^-$ L $\rightarrow$ wire: it shorts the $12\ \Omega$. $i_L = 24/4 = 6\ \text{A}$

0^+ continuity: $i_L(0^+) = 6\ \text{A}$, now forced through the $12\ \Omega$

τ $L/R = 2/12 = 1/6\ \text{s}$ Final: $i_L(\infty) = 0$

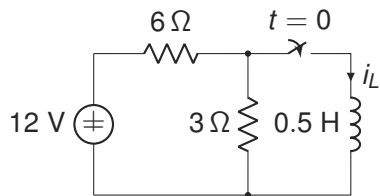
$$i_L(t) \quad i_L(t) = 6 e^{-6t} \text{ A}$$

$v_L(0^+)$ i_L flows up through the $12\ \Omega$: $v_L = -(6)(12) = -72\ \text{V}$

The inductor kick

The inductor current can't jump, but its **voltage** can, and it jumped from 0 to 72 V (three times the source). This is what makes sparks at switches.

Try it: Example 8, RL with Thévenin



$i_L = 0$ before the switch closes at $t = 0$.

Find $i_L(t)$ using the Thévenin equivalent seen by the inductor.

your work here ↓

Example 8: solution

Thévenin seen by L: $V_{Th} = 12 \cdot \frac{3}{6+3} = 4 \text{ V}$, $R_{Th} = 6 \parallel 3 = 2 \Omega$

Initial $i_L(0^+) = 0$

Final L $\rightarrow$ wire: $i_L(\infty) = V_{Th}/R_{Th} = 2 \text{ A}$

$$\tau \quad L/R_{Th} = 0.5/2 = 0.25 \text{ s}$$

$$i_L(t) \quad i_L(t) = 2(1 - e^{-4t}) \text{ A}$$

Shortcut: any single-C or single-L circuit becomes a simple RC or RL loop once you Thévenin everything else.

Second-order circuits (one L and one C)

[NOT on formula sheet: memorize]

Series RLC:

$$\alpha = \frac{R}{2L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Characteristic equation:

$$s^2 + 2\alpha s + \omega_0^2 = 0$$

Parallel RLC:

$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

α is how fast energy is burned (damping). ω_0 is how fast energy sloshes between L and C. The comparison decides the shape.

Where the characteristic equation comes from

Series RLC, no source, current i . KVL, then differentiate once and divide by L :

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0$$

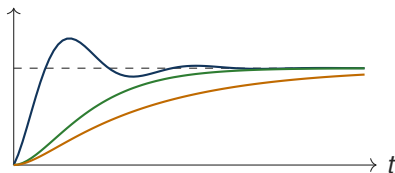
Try $i = Ae^{st}$ (its derivatives just multiply it by s):

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 = s^2 + 2\alpha s + \omega_0^2 \Rightarrow \alpha = \frac{R}{2L}, \omega_0 = \frac{1}{\sqrt{LC}}$$

The two roots decide everything: real and different (overdamped), equal (critical), or complex $-\alpha \pm j\omega_d$, which gives the cos and sin terms (underdamped).

The three cases

Case	Condition	Natural response
Overdamped	$\alpha > \omega_0$	$A_1 e^{s_1 t} + A_2 e^{s_2 t}$ (two real roots)
Critically damped	$\alpha = \omega_0$	$(A_1 + A_2 t) e^{-\alpha t}$
Underdamped	$\alpha < \omega_0$	$e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$



underdamped (overshoots)
critical (fastest, no overshoot)
overdamped (slow)

$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ is the damped frequency.

Try it: Example 9, classify the response

A series RLC has $L = 10 \text{ mH}$ and $C = 1 \text{ } \mu\text{F}$.

(a) Find ω_0 .

(b) For $R = 100 \text{ } \Omega$, $200 \text{ } \Omega$ and $500 \text{ } \Omega$: find α , the case, and the roots (or ω_d).

your work here ↓

Example 9: solution

$$(a) \quad \omega_0 = 1/\sqrt{(10^{-2})(10^{-6})} = 10^4 \text{ rad/s}$$

$$R = 100 \quad \alpha = 100/0.02 = 5000 < \omega_0: \text{ underdamped}$$

$$\omega_d = \sqrt{10^8 - 2.5 \times 10^7} = 8660 \text{ rad/s}$$

$$R = 200 \quad \alpha = 10\,000 = \omega_0: \text{ critically damped}, \quad s = -10^4 \text{ (double)}$$

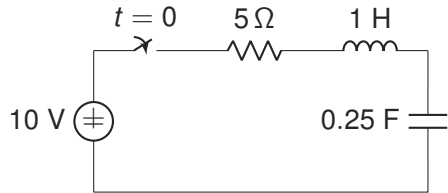
$$R = 500 \quad \alpha = 25\,000 > \omega_0: \text{ overdamped}$$

$$s = -25\,000 \pm \sqrt{6.25 \times 10^8 - 10^8} = -2087, -47\,913$$

Shortcut: critical resistance for series RLC is $R = 2\sqrt{L/C}$. Here

$$2\sqrt{0.01/10^{-6}} = 200 \, \Omega \checkmark$$

Try it: Example 10, second-order step



Everything starts at zero ($v_C = 0$, $i_L = 0$). Find $v_C(t)$ for $t > 0$.

your work here ↓

Example 10: solution (1/2)

Parameters $\alpha = \frac{5}{2(1)} = 2.5$, $\omega_0 = \frac{1}{\sqrt{0.25}} = 2 \Rightarrow$ overdamped

Roots $s = -2.5 \pm \sqrt{6.25 - 4} = -2.5 \pm 1.5$: $s_1 = -1$, $s_2 = -4$

Final $C \rightarrow$ gap: $v_C(\infty) = 10 \text{ V}$

Form $v_C(t) = 10 + A_1 e^{-t} + A_2 e^{-4t}$

Two initial conditions are needed

$v_C(0^+) = 0$, and $\frac{dv_C}{dt}(0^+) = \frac{i_C(0^+)}{C} = \frac{i_L(0^+)}{C} = 0$ (series: same current).

Example 10: solution (2/2)

Derivative differentiate the form: $v_C'(t) = -A_1 e^{-t} - 4A_2 e^{-4t}$, so at $t = 0$: $v_C'(0) = -A_1 - 4A_2$

$$v_C(0) = 0 \quad 10 + A_1 + A_2 = 0 \quad (\text{every exponential equals 1 at } t = 0)$$

$$v_C'(0) = 0 \quad -A_1 - 4A_2 = 0 \Rightarrow A_1 = -4A_2$$

Solve substitute: $10 - 4A_2 + A_2 = 0 \Rightarrow A_2 = \frac{10}{3}, A_1 = -\frac{40}{3}$

Result $v_C(t) = 10 - \frac{40}{3}e^{-t} + \frac{10}{3}e^{-4t} \text{ V}$

Check $t = 0: 10 - 13.33 + 3.33 = 0 \checkmark \quad t \rightarrow \infty: 10 \checkmark$

Underdamped: overshoot, peak, settling

[NOT on formula sheet: memorize]

For an underdamped step response (it overshoots, then rings down):

$$x(t) = x(\infty) + e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

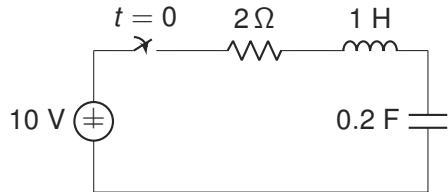
Peak time $t_p = \pi / \omega_d$ (first peak)

Overshoot (zero initial conditions) $e^{-\alpha \pi / \omega_d} \times 100\%$ of the final value

Settling time (2%) $t_s \approx 4 / \alpha$ (rule of thumb)

Shortcut: α sets how fast the ringing **dies** (envelope $e^{-\alpha t}$); ω_d sets how fast it **oscillates**. More series R means more α : less overshoot.

Try it: Example 11, underdamped step



Everything starts at zero. Find $v_C(t)$, the peak time, the peak value, the overshoot and the settling time.

your work here ↓

Example 11: solution (1/3)

Parameters $\alpha = R/2L = 2/2 = 1$, $\omega_0 = 1/\sqrt{LC} = 1/\sqrt{0.2} = 2.236$:
 $\alpha < \omega_0$, underdamped

$$\omega_d \sqrt{\omega_0^2 - \alpha^2} = \sqrt{5 - 1} = 2 \text{ rad/s}$$

Form final value 10 V (C $\rightarrow$ gap):
 $v_C = 10 + e^{-t}(A_1 \cos 2t + A_2 \sin 2t)$

$$v_C(0) = 0 \quad \cos 0 = 1, \sin 0 = 0: 10 + A_1 = 0 \Rightarrow A_1 = -10$$

Example 11: solution (2/3)

Derivative product rule at $t = 0$: $v_C'(0) = -\alpha A_1 + \omega_d A_2$

$$v_C'(0) = 0 \quad i_L(0) = 0 \text{ and series, so } i_C(0) = 0: 10 + 2A_2 = 0 \Rightarrow A_2 = -5$$

Result

$$v_C(t) = 10 - e^{-t}(10 \cos 2t + 5 \sin 2t) \text{ V}$$

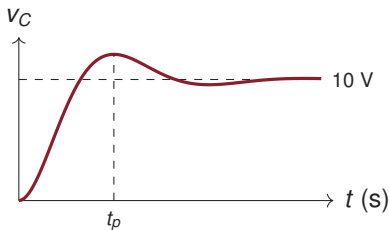
Example 11: solution (3/3)

Peak time $t_p = \pi/2 = 1.57 \text{ s}$

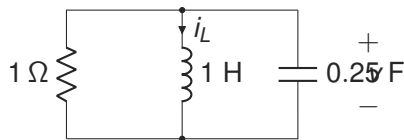
Peak $10 + 10e^{-1.571} = 12.08 \text{ V}$

Overshoot $e^{-\pi/2} = 20.8\%$

Settling $\approx 4/\alpha = 4 \text{ s}$ (exact: 3.7 s)



Try it: Example 12, parallel RLC



No source. At $t = 0$: $v(0) = 10\ \text{V}$ and $i_L(0) = 0$.
Find $v(t)$.

your work here ↓

Example 12: solution

Parameters parallel: $\alpha = \frac{1}{2RC} = \frac{1}{2(1)(0.25)} = 2$, $\omega_0 = \frac{1}{\sqrt{0.25}} = 2$

Case $\alpha = \omega_0$: critically damped, $v = (A_1 + A_2 t)e^{-2t}$

$v(0)$ $A_1 = 10$

$v'(0)$ KCL: $Cv' = -(v/R + i_L) \Rightarrow v'(0) = -\frac{10 + 0}{0.25} = -40$

Solve $v' = A_2 - 2A_1$ at $t = 0$: $A_2 - 20 = -40 \Rightarrow A_2 = -20$

Result $v(t) = (10 - 20t)e^{-2t} \text{ V}$

The second initial condition

Never assume the slope is zero

The second initial condition always comes from the circuit at $t = 0^+$:

$v'_C(0^+) = i_C(0^+)/C$, with $i_C(0^+)$ from KCL.

$i'_L(0^+) = v_L(0^+)/L$, with $v_L(0^+)$ from KVL.

In Example 10 and Example 11 the slope happened to be zero because the circuits started fully at rest. In Example 12 it was -40 V/s.

Traps

Check these before you box an answer

- Capacitors combine opposite to resistors (parallel adds).
- Continuity is for v_C and i_L only. i_C , v_L and resistor values **can** jump.
- τ uses R_{Th} of the circuit **after** the switch, seen by the C or L.
- $\tau = RC$ but $\tau = L/R$. Don't write RL .
- Watch prefixes: $\mu F \times k\Omega = ms$.
- Second order needs **two** initial conditions: the value and its slope.

Exam-format traps

DC questions with L and C

“Find the energy stored” with a DC source: it’s a steady-state question. Don’t write differential equations.

Give the full expression

A transient answer should be a function of time with units, like $v_C(t) = 8e^{-5t}$ V for $t > 0$.

Show each step

Write $x(0^+)$, $x(\infty)$ and τ separately before the final formula. Each is usually worth marks.

Summary (1/2)

Capacitor	$i = C dv/dt$, $W = \frac{1}{2}Cv^2$, DC: open, v_C can't jump
Inductor	$v = L di/dt$, $W = \frac{1}{2}Li^2$, DC: wire, i_L can't jump
Combining	L like R; C opposite (parallel adds, series product over sum)
DC steady state	C $\rightarrow$ gap, L $\rightarrow$ wire, solve resistively

Summary (2/2)

First order	$x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$, $\tau = R_{Th}C$ or L/R_{Th}
Recipe	0^- steady state $\rightarrow$ continuity $\rightarrow \infty$ steady state $\rightarrow \tau \rightarrow$ formula
Timing	63% at 1τ , done at 5τ
Second order	α (series $R/2L$, parallel $1/2RC$), $\omega_0 = 1/\sqrt{LC}$; over / critical / under

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: capacitor

A $2\ \mu\text{F}$ capacitor's voltage rises linearly from 0 to 6 V in 3 ms, then stays at 6 V.

(a) Current during the ramp? (b) Current after 3 ms? (c) Final stored energy?

your work here ↓

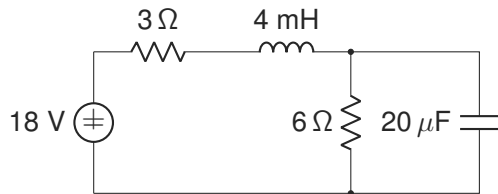
P2: combining

(a) A $2\ \mu\text{F}$ and a $4\ \mu\text{F}$ capacitor are in parallel, and that pair is in series with a $3\ \mu\text{F}$. Find C_{eq} .

(b) A $6\ \text{mH}$ and a $3\ \text{mH}$ inductor are in series, and that pair is in parallel with an $18\ \text{mH}$. Find L_{eq} .

your work here ↓

P3: DC steady state



DC for a long time. Find i_L , v_C , W_L and W_C .

your work here ↓

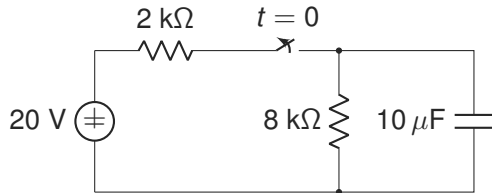
P4: RC timing

An uncharged $10\ \mu\text{F}$ capacitor charges through $4\ \text{k}\Omega$ from a $12\ \text{V}$ source, starting at $t = 0$.

(a) τ ? (b) $v_C(t)$? (c) How long until $v_C = 9\ \text{V}$?

your work here ↓

P5: RC discharge

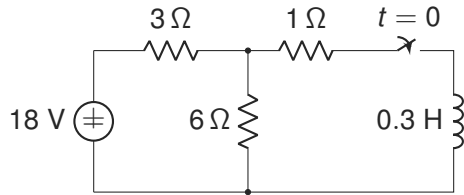


The switch was closed a long time and opens at $t = 0$.

Find $v_C(t)$ and v_C at 80 ms.

your work here ↓

P6: RL with Thévenin



$i_L = 0$ before $t = 0$. Find $i_L(t)$.

your work here ↓

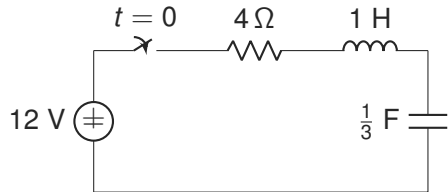
P7: classify an RLC

A series RLC has $L = 20 \text{ mH}$, $C = 5 \mu\text{F}$, $R = 40 \Omega$.

(a) ω_0 and α . (b) Which case? (c) ω_d . (d) What R would make it critically damped?

your work here ↓

P8: second-order step response



Everything starts at zero. Find $v_C(t)$ for $t > 0$.

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

Plan capacitor current = $C \times$ slope of its voltage

(a) slope during the ramp: $\frac{\Delta v}{\Delta t} = \frac{6 \text{ V}}{3 \text{ ms}} = 2000 \text{ V/s}$

(a) current $i = C \frac{dv}{dt} = (2 \times 10^{-6})(2000) =$ 4 mA

(b) after 3 ms the voltage is flat, slope = 0, so $i = 0$

(c) energy depends only on the final voltage:

$$W = \frac{1}{2} C v^2 = \frac{1}{2} (2 \times 10^{-6}) (6^2) =$$
 $36 \mu\text{J}$

P2: solution

Rule capacitors combine **opposite** to resistors; inductors combine **like** resistors

(a) parallel parallel capacitors add: $2 + 4 = 6 \mu\text{F}$

(a) series two capacitors in series: product over sum, $\frac{6 \times 3}{6 + 3} =$

$$2 \mu\text{F}$$

(b) series series inductors add: $6 + 3 = 9 \text{ mH}$

(b) parallel two inductors in parallel: product over sum, $\frac{9 \times 18}{9 + 18} =$

$$6 \text{ mH}$$

P3: solution

Plan DC for a long time: steady state, so replace L with a wire and C with a gap, then solve the resistive circuit

Circuit the C branch carries nothing, so one loop: 18 V, 3 Ω , (wire), 6 Ω : total 9 Ω

i_L the inductor is in that loop: $i_L = 18/9 = 2 \text{ A}$

v_C the capacitor is in parallel with the 6 Ω , so it has the same voltage: $2 \times 6 = 12 \text{ V}$

W_L $\frac{1}{2}Li_L^2 = \frac{1}{2}(4 \times 10^{-3})(2^2) = 8 \text{ mJ}$

W_C $\frac{1}{2}Cv_C^2 = \frac{1}{2}(20 \times 10^{-6})(12^2) = 1.44 \text{ mJ}$

P4: solution (1/2)

Values $v_C(0^+) = 0$ (uncharged), $v_C(\infty) = 12 \text{ V}$ (C $\rightarrow$ gap, no current, no drop on R)

(a) $\tau = RC = (4000)(10 \times 10^{-6}) = 40 \text{ ms}$

(b) formula: $v_C = 12 + (0 - 12)e^{-t/\tau} \Rightarrow$
$$v_C(t) = 12(1 - e^{-t/0.04}) \text{ V}$$

P4: solution (2/2)

(c) setup set $v_C = 9$: $9 = 12(1 - e^{-t/\tau}) \Rightarrow e^{-t/\tau} = 1 - \frac{9}{12} = 0.25$

(c) solve take $\ln$ of both sides: $-t/\tau = \ln 0.25 = -1.386 \Rightarrow$
 $t = 1.386 \tau = 55.5 \text{ ms}$

Engineering shortcut: “how long to reach X ”

$$t = \tau \ln \frac{x(0) - x(\infty)}{x(t) - x(\infty)} \quad \text{here: } 40 \ln \frac{0 - 12}{9 - 12} = 40 \ln 4 = 55.5 \text{ ms}$$

P5: solution

$t = 0^-$ closed for a long time: C $\rightarrow$ gap. The 2 k and 8 k form a divider: $v_C = 20 \cdot \frac{8}{2+8} = 16 \text{ V}$

$t = 0^+$ continuity: $v_C(0^+) = 16 \text{ V}$

After the switch opens and disconnects the source and the 2 k: the C discharges through the 8 k alone, so $v_C(\infty) = 0$

τ resistance seen by the C is 8 k: $\tau = (8000)(10 \times 10^{-6}) = 80 \text{ ms}$

$$v_C(t) = 0 + (16 - 0)e^{-t/\tau}: \quad v_C(t) = 16 e^{-t/0.08} \text{ V}$$

80 ms $t = \tau$, so $16 e^{-1} = 5.89 \text{ V}$ (36.8% left)

P6: solution

Plan replace everything the inductor sees by its Thévenin equivalent, then it's a simple RL loop

V_{Th} remove L: open circuit, no current in the $1\ \Omega$; the 18 V divides across 3 and 6: $18 \cdot \frac{6}{3+6} = 12\text{ V}$

R_{Th} kill the 18 V: $3 \parallel 6 = 2$, plus the series $1\ \Omega$: $R_{Th} = 3\ \Omega$

Values $i_L(0^+) = 0$, $i_L(\infty) = V_{Th}/R_{Th} = 4\text{ A}$ (L $\rightarrow$ wire),
 $\tau = L/R_{Th} = 0.3/3 = 0.1\text{ s}$

Result $i_L = 4 + (0 - 4)e^{-t/0.1}$: $i_L(t) = 4(1 - e^{-10t})\text{ A}$

P7: solution

(a) ω_0 $1/\sqrt{LC} = 1/\sqrt{(0.02)(5 \times 10^{-6})} = 1/\sqrt{10^{-7}} = 3162 \text{ rad/s}$

(a) α series RLC: $R/2L = 40/(2 \times 0.02) = 1000 \text{ /s}$

(b) $\alpha < \omega_0$: underdamped (it will overshoot and ring)

(c) $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{10^7 - 10^6} = 3000 \text{ rad/s}$

(d) critical needs $\alpha = \omega_0$: $R/2L = 1/\sqrt{LC} \Rightarrow$
 $R = 2\sqrt{L/C} = 2\sqrt{4000} = 126.5 \Omega$

P8: solution (1/2)

Parameters $\alpha = R/2L = 4/2 = 2$, $\omega_0^2 = 1/LC = 1/(1 \cdot \frac{1}{3}) = 3$:
 $\alpha^2 = 4 > 3$, overdamped

Roots $s^2 + 4s + 3 = (s + 1)(s + 3) = 0$: $s = -1, -3$

Form final 12 V (C $\rightarrow$ gap): $v_C = 12 + A_1 e^{-t} + A_2 e^{-3t}$, so
 $v'_C(0) = -A_1 - 3A_2$

P8: solution (2/2)

Conditions $v_C(0) = 0$: $12 + A_1 + A_2 = 0$; $v'_C(0) = i_L(0)/C = 0$:
 $-A_1 - 3A_2 = 0$

Solve $A_1 = -3A_2$, then $12 - 2A_2 = 0$: $A_2 = 6$, $A_1 = -18$

Result $v_C(t) = 12 - 18e^{-t} + 6e^{-3t}$ V (check $t = 0$:
 $12 - 18 + 6 = 0$ ✓)